

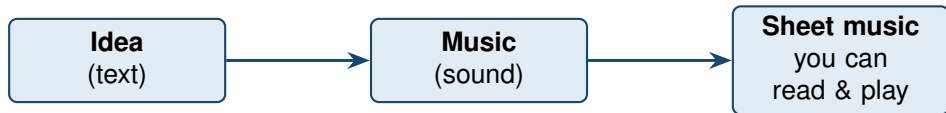
From Idea to Sheet Music

Two halves of one music-AI system



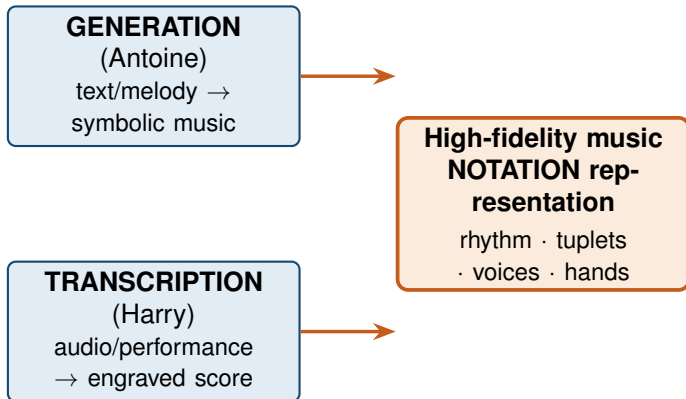
Harry Ilanyan • Antoine • Bayen Lab

The dream



Type *"a melancholic Chopin-style nocturne"* → get **audio** and a clean, playable **score**.

The lab on one slide



Different inputs, same core problem.

ACT 1

Generating music

Teaching a model to compose like a pianist

- **Goal:** given a short prompt (a melody, a few bars), **generate an idiomatic piano continuation / harmonization**.
- **Hard because:** piano music is **two hands, polyphonic**, with ornaments, dynamics, pedaling, rubato — not a single melody line.



CWT = Compound-Word Transformer

Key idea: every musical event = one **compound token** of **14 parallel heads**, each a different musical dimension.

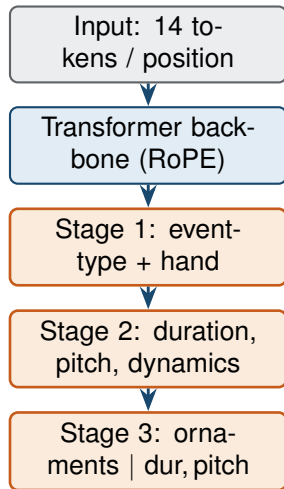
Type	Duration	Pitch(+oct+acc)	Ornament
Articulation	Tie	Slur	Phrase
Voicing	Beam	Hand (RH/LH)	Dynamic
Pedal	Tempo		

Trained on **Humdrum **kern** (compact text notation; the KernScores tradition).

[Tech, optional ~10s] 14 categorical heads predicted jointly; total vocab only a *few hundred* tokens vs 20,000+ for raw kern $\Rightarrow$ whole pieces fit in a 2,048-token window.

The architecture

- **~16.2M-param** Transformer, 10 layers, **RoPE** positions.
- **3-stage cascaded decoding** (models conditional dependencies between attributes).
- **Hand-aware:** declares RH vs LH at every step.
- v1 $\rightarrow$ v2 (training now) $\rightarrow$ v3 (atomic chords).



[Tech, optional ~10s] Stage 3 predicts ornaments *conditioned* on the decided duration & pitch; embedding is `concat(14×320)→320`.

~219,000 piano scores

- **Sources:** PDMX + ASAP + Chopin first editions + **KernScores**.
- **Coverage:** >99.5% of pieces fit in a 2,048-token window (the model sees whole pieces).
- **Honest status:** v2 is mid-training (epoch 2, loss ~ 0.076); v1 already produces coherent 6-bar continuations.

LIVE DEMO 1

Antoine generates music

(6-bar Chopin prompt → AI continuation, played in MuseScore)

Demo steps + fallback in `LAB_PRESENTATION_SCRIPT.md`

Where Antoine's half stands

OK compound tokenizer, v1 generation, kern $\rightarrow$ MusicXML $\rightarrow$ MuseScore pipeline.

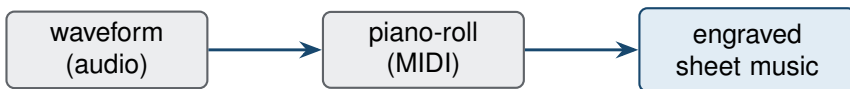
WIP v2 training (pedal/tempo/3-stage), held-out eval set, multi-staff (3+ voices).

Next style control (per-composer), longer-range coherence.

ACT 2

Reading music

From a performance to a readable score



“What was played” (messy, expressive timing) → *“what a musician reads”* (quantized rhythm, beams, voices, hands).

Two blocks



- 4 transcribers integrated; **hFT-Transformer** is best (**96.7% onset-F1**).
- Block 2 = the Beyer & Dai model (ISMIR 2024) — the engine **Songscription** is built on.

[Tech, optional ~10s] a path hands the transcriber's float-precision notes *straight* into the score model — no MIDI-file re-quantization.

Result #1: reproduced SOTA — to within noise

MUSTER (lower=better)	Ours	Paper (Beyer & Dai 2024)
MeanER	11.18	11.30
PitchER	3.19	3.11
MissRate	7.83	7.56
OnsetER	14.50	15.55
OffsetER	23.73	23.84

MUSTER MeanER = the standard score-transcription error rate. All deltas $< 0.12 \Rightarrow$ **validates the whole eval harness.**

[Tech, optional ~10s] MUSTER is a C++ aligner over two MusicXMLs (scores *notation*); on Apple Silicon the MPS backend is degenerate $\Rightarrow$ eval runs on CPU.

Result #2: on clean piano, commercial quality

Clean piano: end-to-end MeanER $\approx$ **2.75**



Honest caveat

2.75 is the **easy, in-distribution end** — *not* a like-for-like win over the product: the score model is the same open checkpoint Songscription runs. The apples-to-apples number is the **11.18 reproduction**. The hard Romantic repertoire is next.

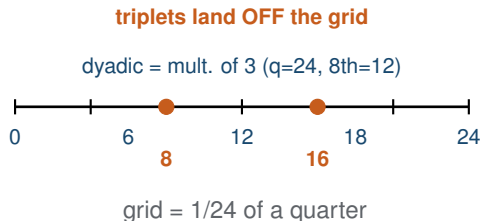
Where it breaks: triplet rhythm

- **Not** the transcriber: *perfect* GT MIDI into Block 2 fails the same way.
- **Not** pitch: the notes are right. It's **where they sit in time** — triplets/tuplets.

Liszt “Mazeppa” (~49% triplets):

tuplets produced $\approx$ **half** of GT $\Rightarrow$ wrong meters $\Rightarrow$ **70%** too many measures $\Rightarrow$ alignment collapses.

[Tech, optional ~10s] triplets = non-multiple-of-3 buckets (8, 16); individually *rare* $\Rightarrow$ the model learns to never bet on them.



Why is tuplet placement so hard? A careful answer

Diagnostic: the model places **0%** of notes at triplet positions on hard pieces (GT $\approx$ 4.8%).

Lever tried	Triplet placement	MUSTER
Inference cleanup (A2 + pad + per-head)	n/a	11.79 $\rightarrow$ 11.41 (win)
Corpus reshape ($\gamma=0.5-1.5$)	0% $\rightarrow$ 19.9% (over)	regresses
Self-training (pseudo-label MAESTRO)	0.9%	regresses
Loss weighting (tuplet-weight $\times 5$)	0%	over-emits
Beat-relative representation (B2)	0%	$\approx$ baseline
Rendered paired (24.73% triplet targets)	0%	undertrained
From scratch (AR undertrained; non-AR matched val)	0%	—

No lever hit the *correct* rate. Matched the released model's val loss + within ~ 1 MUSTER pt — but never its placement. **Leading explanation: the data** (our corpus is $\sim 1.7\%$ triplets). The decisive converged-on-tuplet-rich run is *open* $\Rightarrow$ “not recovered within our budget,” *not* “impossible.”

[Tech, optional $\sim 10s$] B2 recoding (within-quarter + quarter-index) is lossless & concentrates 100% of triplets into shared buckets — still 0% learned: representation *necessary, not sufficient*.

LIVE DEMO 2

Harry transcribes a performance

(clean Chopin clip → hFT → score PDF)

Demo steps + fallback PDF in `LAB_PRESENTATION_SCRIPT.md` /
`benchmark/chopin_op10/transformer/`

ACT 3

The convergence

Same idea, both halves: multi-head compound tokens

Antoine (generation)

14 parallel heads

type · duration · pitch
· ornament · **hand** ·
dynamics · pedal ...

Harry (transcription)

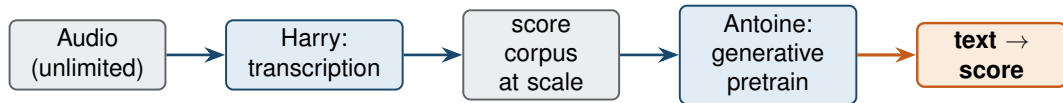
13 parallel streams

offset · duration · downbeat
· pitch · accidental ·
voice · stem · **hand** ...

Two people, two directions, independently chose to represent music as a set of **parallel attribute streams**. Not a coincidence — it's the right abstraction for notation.

[Tech, optional ~10s] Deeper still: **both models are RoFormers (RoPE)**. Compound tokens *and* rotary encoding, arrived at independently.

One system: the data engine + the generator



- Scores are scarce; audio is unlimited. **Transcription manufactures training data for generation.**
- Both stages share **one notation representation** — owning it is the lab's real moat.

Division of labor that actually composes

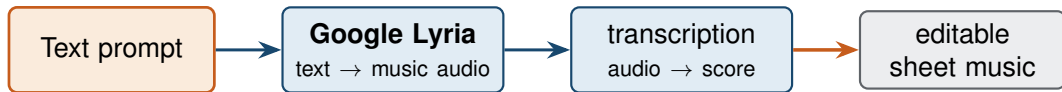
- **Antoine:** the **generator** (text/melody $\rightarrow$ music) + the symbolic-music modeling.
- **Harry:** the **data engine** (audio $\rightarrow$ score at scale) + **the notation-representation / fidelity** both models depend on.

Harry's role isn't "the transcription guy" — it's the **owner of the shared notation representation**.

ACT 4

The full loop

Text → Music → Score — the whole loop



A concept demo of the end goal, stitched from today's best pieces — Lyria for generation, our transcription for notation.

LIVE DEMO 3

The full loop: text → audio → score
(text → Lyria audio → transcription → sheet music)

[implementation to confirm; recommend pre-recording]

Framing: “Demo 1 = Antoine’s real generator. Demo 2 = my real transcriber. Demo 3 = the vision —
where our two real systems slot in.”

Where this goes (Bayen's direction)

1. **Data engine:** transcription manufactures score corpora to pretrain the generator (use the released model as the placement-capable engine; filter dense-tuplet cases).
2. **Shared notation representation:** converge our 13/14-head schemes; evaluate a **rhythm-honest, LLM-friendly format (e.g. ABC)** both halves use — and that makes *text* conditioning natural.
3. **Close the loop:** replace Lyria/Songscription in the demo with Antoine's generator + Harry's pipeline $\Rightarrow$ a true **text** $\rightarrow$ **score** system the lab owns.

Takeaways

- Two halves — **generation (Antoine)** + **transcription (Harry)** — both working end-to-end with live demos.
- They're **one system**: same representation idea (compound multi-head tokens + RoPE), same hard problem (notation fidelity), composing into **text** → **score**.
- Real results: reproduced SOTA (**11.18**, within noise), commercial quality on clean piano (**2.75**), a rigorous characterization of the tuplet ceiling (a *data* problem, not a dead end).
- **The ask:** _____ [compute for the data engine / shared-representation decision / green-light to merge tracks]

Thank you

Questions — and more demos

BACKUP SLIDES

anticipated Q&A + technical deep-dive — pull up only if asked

Q&A: “Your 2.75 vs 11 — isn’t that cherry-picked?”

Yes, deliberately, and we flag it: 2.75 is the clean in-distribution end; the 11 is the full-test average. The honest, apples-to-apples number is the **11.18 reproduction**. 2.75 shows the *ceiling* of what’s possible on the music it’s designed for; the tuple finding shows the *floor* on hard music.

Q&A: “Did you just not train long enough?”

Honest answer: partly — our capstone from-scratch run was **undertrained**, so we can't claim impossibility. What we *can* say is sharper: a model that matched the released *validation loss* still showed zero correct-rate placement, and pushing it with data, representation, and loss never hit the correct rate. Leading explanation: the **data** (corpus $\sim 1.7\%$ tuples). So it's “not recovered within our compute/data budget; the data is the binding constraint” — pointing to a concrete fix (tuple-rich corpus, trained to convergence). One cheap step not yet done: email the paper's author.

Q&A: novelty vs. off-the-shelf; data engine real?

Novel: (1) the rigorous, controlled characterization of the tuplet-placement ceiling (unpublished); (2) the data-engine framing (turn unlimited audio into scarce score data); (3) the convergence insight (generation + transcription share one representation).

Data engine real? The pieces are real — a transcription pipeline that reproduces SOTA + a render/pairing pipeline that already built a 23,000-piece paired corpus in minutes. Aspirational: scaling it and wiring it into pretraining — that's the ask.

D1 — The two tokenizers, side by side

Antoine — CWT, 14 heads

Type, Duration, Pitch(+oct+acc), Ornament, Articulation, Tie, Slur, Phrase, Voicing, Beam, **Hand**, Dynamic, Pedal, Tempo.

Vocab $\approx$ few hundred; `concat(14  $\times$  320)  $\rightarrow$  320`; **RoPE**;
3-stage decode; ~ 16.2 M params; autoregressive.

Harry — MIDI2ScoreTransformer, 13 streams + pad

offset, downbeat, duration, pitch, accidental, keysignature, velocity, grace, trill, staccato, voice, stem, **hand**, + pad gate.

RoFormer enc-4/dec-4, hidden 512, 8 heads, ~ 32.6 M params; non-autoregressive (mask-predict).

Shared, unplanned: compound multi-head tokens + RoPE.

D2 — MUSTER, in detail

Python wrapper around a **C++ aligner** over two MusicXMLs. Six error rates — PitchER, MissRate, ExtraRate, OnsetER, OffsetER → **MeanER** (their mean). Plus engraving sub-metrics (note spelling, voice assignment, stem direction, staff assignment) + voice precision/recall/F1. **Aborts on 3-staff scores** (limits OOD eval). Our 11.18 matched the paper on *all five* sub-metrics within 0.12.

D3 — The 1/24 grid + the placement diagnostic

Grid = **1/24 of a quarter**. Dyadic = multiples of 3 (quarter 24, eighth 12, 16th 6); triplet/sextuplet = off-multiples (8, 16, 4, 20). `diag_offset_phase.py` measures the fraction of onsets at triplet phases: **GT $\approx$ 4.78%, our models 0%**.

B2 = within-quarter offset + integer `quarter_idx`. Verified: lossless round-trip, 100% triplet concentration into shared buckets, non-B2 byte-identical. Conclusion: representation **necessary, not sufficient**.

D4 — The exhaustive placement matrix + data engine

Inference cleanup → 11.79→**11.41** (only robust gain); global logit-shift → over-emits; corpus reshape → **19.9%** (over-shoots, regresses); self-training → 0.9%; tuple-weight $\times 5$ → 0%; B2 → 0%; paired distillation (2.87% targets) → 0%; rendered paired (**24.73%** targets) → 0% (undertrained); from-scratch AR (val 1.3, undertrained) + non-AR (matched val 0.51) → 0%.

Data engine: 23,780 PDMX scores rendered → valid per-beat-aligned paired data in ~11 min; tuple-rich subset 24.73% placement.

D5 — Eval-harness rigor (why the numbers are real)

- **MPS→CPU:** Apple-MPS produced degenerate pad logits $\Rightarrow$ eval forced to CPU.
- **val $\neq$ MUSTER:** re-rank checkpoints by *true* MUSTER, not loss.
- **Leakage audit:** 12/17 eval pieces found in PDMX via *content fingerprints* (transposition-invariant interval n-grams, not titles) $\Rightarrow$ content-dedup tool + blocklist for future training.
- **Pad-threshold bug:** released `generate()` hard-argmaxed pad to binary (any threshold a no-op) — we fixed it (continuous keep-probability + un-zeroed raw streams).